Examiner only

3. Rhys and Elliot have a small piece of an unknown metal. The metal has an **irregular shape**. To identify the metal they find its density and compare the value to known values of the density of common metals.

(a) Describe a method they could use to find the density of the metal. [6 QER]

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(b) The table below gives data on the density of some common metals.

Metal	Density (g/cm^3)
aluminium	2.70
copper	8.96
gold	19.32
iron	7.87
tin	7.26

(i) Rhys and Elliot calculate the density of their metal to be 8.1 g/cm^3.
State which metal the irregular shape is most likely to be. Give a reason for your answer. [2]

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(ii) Rhys and Elliot are not confident that they can correctly identify the metal. Suggest why they think this. [1]

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(iii) The table below shows their results.

Mass (g)	Volume (cm^3)	Density (g/cm^3)
65	8	8.1

Suggest how Rhys and Elliot could get a more accurate value for the density. [1]

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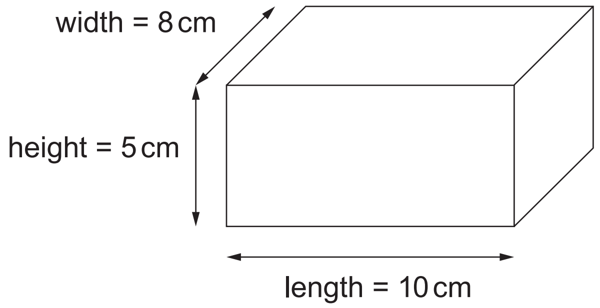
(c) The boys notice that gold has a high density of 19.32 g/cm^3 and they are interested in the mass of a gold block.



(i) Use the equation:

volume = length × height × width

to calculate the volume of the gold block shown above. [1]

Volume = ... cm^3

(ii) Use the equation:

mass = density × volume

to calculate the mass of the gold block. [2]

Mass = ... g

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